

Lecture - 6

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

CORRELATION

Introduction

In today's world we come across many activities, which are dependent on each other. In businesses we see large number of problems involving the use of two or more variables. Identifying these variables and its dependency helps us in resolving the many problems. Many times, there are problems or situations where two variables seem to move in the same direction such as both are increasing or decreasing. At times an increase in one variable is accompanied by a decline in another. For example, family income and expenditure, price of a product and its demand, advertisement expenditure and sales volume etc. If two quantities vary in such a way that movements in one are accompanied by movements in the other, then these quantities are said to be correlated.

Meaning

Correlation is a statistical technique to ascertain the association or relationship between two or more variables. Correlation analysis is a statistical technique to study the degree and direction of relationship between two or more variables.

A correlation coefficient is a statistical measure of the degree to which changes to the value of one variable predict change to the value of another. When the fluctuation of one variable reliably predicts a similar fluctuation in another variable, there's often a tendency to think that means that the change in one causes the change in the other.

Uses of correlations

1. Correlation analysis helps in deriving precisely the degree and the direction of such relationship.
2. The effect of correlation is to reduce the range of uncertainty of our prediction. The prediction based on correlation analysis will be more reliable and near to reality.
3. Correlation analysis contributes to the understanding of economic behaviour, aids in locating the critically important variables on which others depend, may reveal to the economist the connections by which disturbances spread

and suggest to him the paths through which stabilizing forces may become effective

4. Economic theory and business studies show relationships between variables like price and quantity demanded advertising expenditure and sales promotion measures etc.
5. The measure of coefficient of correlation is a relative measure of change.

Types of Correlation:

Correlation is described or classified in several different ways. Three of the most important are:

- i. Positive and Negative
- ii. Simple, Partial and Multiple
- iii. Linear and non-linear

I. Positive, Negative and Zero Correlation:

Whether correlation is positive (direct) or negative (in-versa) would depend upon the direction of change of the variable.

Positive Correlation: If both the variables vary in the same direction, correlation is said to be positive. It means if one variable is increasing, the other on an average is also increasing or if one variable is decreasing, the other on an average is also decreasing, then the correlation is said to be positive correlation.

For example, the correlation between heights and weights of a group of persons is a positive correlation.

Height (cm): X	158	160	163	166	168	171	174	176
Weight (kg): Y	60	62	64	65	67	69	71	72

Negative Correlation: If both the variables vary in opposite direction, the correlation is said to be negative. It means if one variable increases, but the other variable decreases or if one variable decreases, but the other variable increases, then the correlation is said to be negative correlation. For example, the correlation between the price of a product and its demand is a negative correlation.

Price of Product (Rs. Per Unit): X	6	5	4	3	2	1
Demand (In Units): Y	75	120	175	250	215	400

Zero Correlation: Actually, it is not a type of correlation but still it is called as zero or no correlation. When we don't find any relationship between the variables then, it is said to be zero correlation. It means a change in value of one variable doesn't influence or change the value of another variable. For example,

the correlation between weight of person and intelligence is a zero or no correlation.

II. Simple, Partial and Multiple Correlation:

The distinction between simple, partial and multiple correlation is based upon the number of variables studied.

Simple Correlation: When only two variables are studied, it is a case of simple correlation. For example, when one studies relationship between the marks secured by student and the attendance of student in class, it is a problem of simple correlation.

Partial Correlation: In case of partial correlation, one studies three or more variables but considers only two variables to be influencing each other and the effect of other influencing variables being held constant. For example, in above example of relationship between student marks and attendance, the other variable

Multiple Correlation: When three or more variables are studied, it is a case of multiple correlation. For example, in above example if study covers the relationship between student marks, attendance of students, effectiveness of teacher, use of teaching aids etc, it is a case of multiple correlation.

Linear and Non-linear Correlation:

Depending upon the constancy of the ratio of change between the variables, the correlation may be Linear or Non-linear Correlation.

Linear Correlation: If the amount of change in one variable bears a constant ratio to the amount of change in the other variable, then correlation is said to be linear. If such variables are plotted on a graph paper all the plotted points would fall on a straight line. For example: If it is assumed that, to produce one unit of finished product we need 10 units of raw materials, then subsequently to produce 2 units of finished product we need double of the one unit.

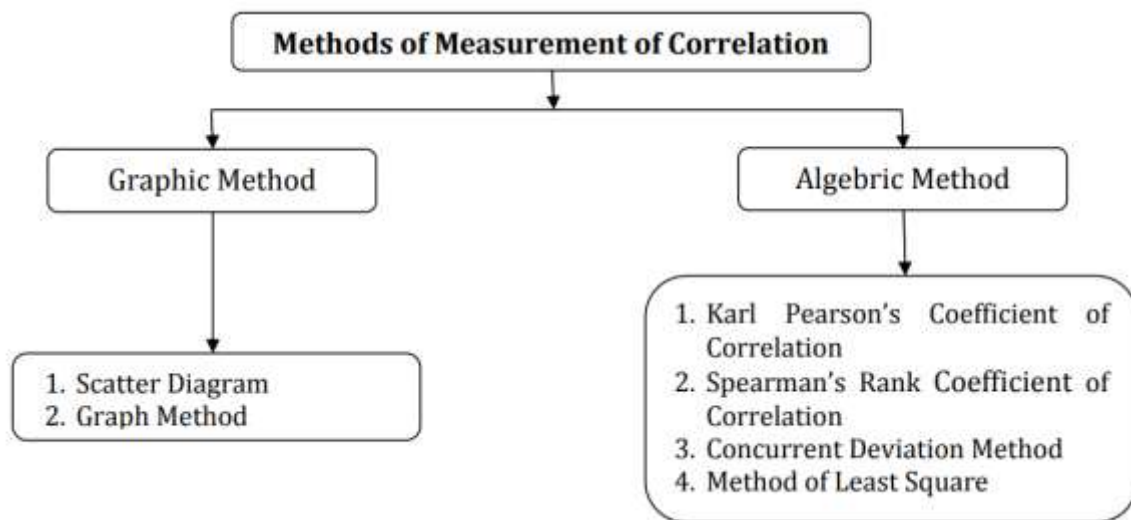
Raw material: X	10	20	30	40	50	60
Finished Product: Y	2	4	6	8	10	12

Non-linear Correlation: If the amount of change in one variable does not bear a constant ratio to the amount of change to the other variable, then correlation is said to be non-linear. If such variables are plotted on a graph, the points would fall on a curve and not on a straight line. For example, if we double the amount of advertisement expenditure, then sales volume would not necessarily be doubled.

Advertisement Expenses: X	10	20	30	40	50	60
Sales Volume: Y	2	4	6	8	10	12

Methods of measurement of correlation:

Quantification of the relationship between variables is very essential to take the benefit of study of correlation. For this, we find there are various methods of measurement of correlation, which can be represented as given below:



Among these methods we will discuss only the following methods:

1. Scatter Diagram
2. Karl Pearson's Coefficient of Correlation
3. Spearman's Rank Coefficient of Correlation

Scatter Diagram

This is graphic method of measurement of correlation. It is a diagrammatic representation of bivariate data to ascertain the relationship between two variables. Under this method the given data are plotted on a graph paper in the form of dot. i.e., for each pair of X and Y values we put dots and thus obtain as many points as the number of observations. Usually, an independent variable is shown on the X-axis whereas the dependent variable is shown on the Y-axis. Once the values are plotted on the graph it reveals the type of the correlation between variable X and Y. A scatter diagram reveals whether the movements in one series are associated with those in the other series.

- **Perfect Positive Correlation:** In this case, the points will form on a straight line falling from the lower left-hand corner to the upper right-hand corner.
- **Perfect Negative Correlation:** In this case, the points will form on a straight line rising from the upper left-hand corner to the lower right-hand corner.
- **High Degree of Positive Correlation:** In this case, the plotted points fall in a narrow band, wherein points show a rising tendency from the lower left-hand corner to the upper right-hand corner.
- **High Degree of Negative Correlation:** In this case, the plotted points fall in a narrow band, wherein points show a declining tendency from upper left-hand corner to the lower right-hand corner.

- Low Degree of Positive Correlation: If the points are widely scattered over the diagrams, wherein points are rising from the left-hand corner to the upper right-hand corner.
- Low Degree of Negative Correlation: If the points are widely scattered over the diagrams, wherein points are declining from the upper left-hand corner to the lower right-hand corner.
- Zero (No) Correlation: When plotted points are scattered over the graph haphazardly, then it indicates that there is no correlation or zero correlation between two variables.